EECS 245, Fall 2026

LEC 14 Inverses and Linear Transformations

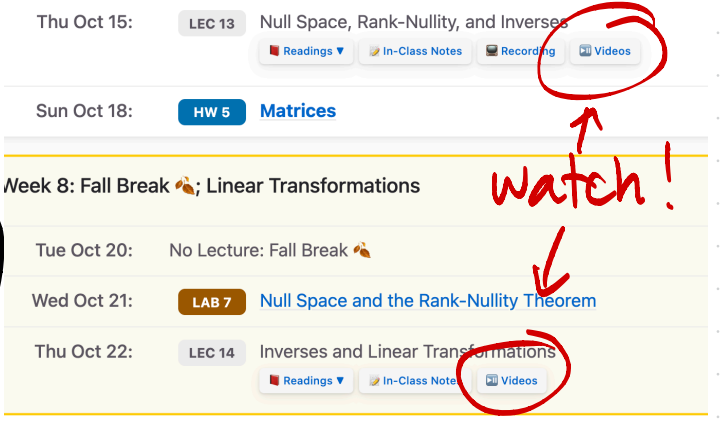
→ Read: Ch. 6.1-6.2

Agenda

- The inverse of a square matrix, A^{-1}
  - What is it?
  - How is it useful?
- Linear transformations
  - Will help us understand **when** a matrix is invertible

Announcements

- HW 6 released tomorrow, due **next** Sunday (11/1)
- Review HW 5, Lab 7 solutions



Inverses?

$(2)^{-1} = \frac{1}{2}$

$(2)^{-1}2x = (2)^{-1}5$

$1x = \frac{5}{2}$

1 is the "identity" element

A's inverse, A^{-1}, should satisfy

$$AA^{-1} = I = A^{-1}A$$

this requires A to be square!

Only square matrices have an inverse

Suppose A is $n \times d$

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$$

$$A\vec{x} = \vec{b}$$

n equations, d unknowns

$\Rightarrow$ this is asking us to write $\vec{b}$ as LC of A's cols

$\Rightarrow$ if there is $\checkmark$ solution, then $\vec{b}$ is in colsp(A).

at least one

Suppose A is $n \times n$.

$$A\vec{x} = \vec{b}$$

$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ unknowns

system of n equations, n unknowns

If A is invertible (means A^{-1} exists), then

$$A^{-1}A\vec{x} = A^{-1}\vec{b}$$

$A^{-1}A = I$

$$\vec{x} = A^{-1}\vec{b}$$

tells me how to make $\vec{b}$ as a linear combination of A's cols

Linear transformation : a type of function

Suppose T is a function, $T: \mathbb{R}^d \to \mathbb{R}^n$

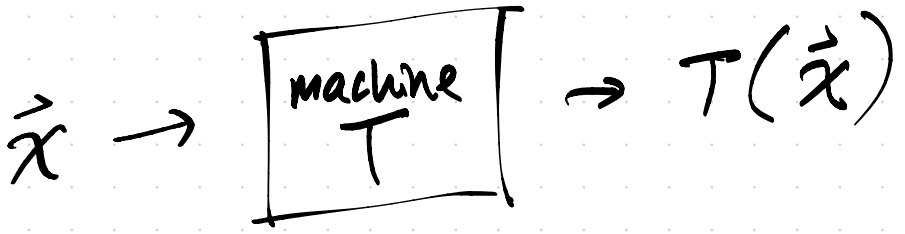
$\mathbb{R}^d$: domain $\quad$ $\mathbb{R}^n$: codomain

T is a linear transformation if it satisfies two properties:

① $T(\vec{x}+\vec{y}) = T(\vec{x}) + T(\vec{y})$ for all $\vec{x}, \vec{y} \in \mathbb{R}^d$

② $T(c\vec{x}) = cT(\vec{x})$ for all $\vec{x} \in \mathbb{R}^d$, $c \in \mathbb{R}$

① + ② : $T(a\vec{x}+b\vec{y}) = a\,T(\vec{x}) + b\,T(\vec{y})$


$\vec{x}$
machine
T
$T(\vec{x})$

Example: $f(x) = x^2$

Linear transformation?

$f(1+2) = f(3) = 9$ $\quad 9 \neq 5$, so no!

$f(1) + f(2) = 1 + 4 = 5$

$f(cx) = c^2 f(x)$ also no!

"affine transformation"

Example : $f(x) = 2x + 5$

makes this not a LT

Linear transformation?

$$f(1+2) = f(3) = 2(3) + 5 = 11$$

$$f(1) + f(2) = 2(1) + 5 + 2(2) + 5 = 16$$

$16 \neq 11$, so no!

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necessary but not sufficient: for all linear transformations,

$$T(\vec{0}) = \vec{0}$$

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 \\ 2x_1 \\ 3x_1 - 4x_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 0 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$\in \mathbb{R}^2$ $\quad \in \mathbb{R}^3$ $\quad A$ $\quad \vec{x}$

$$T(\vec{x}) = A\vec{x}$$

$$T(\vec{x}+\vec{y}) = A(\vec{x}+\vec{y}) = A\vec{x} + A\vec{y} = T(\vec{x}) + T(\vec{y})$$

$$T(c\vec{x}) = A(c\vec{x}) = cA\vec{x} = cT(\vec{x})$$

Big idea: all linear transformations are matrix-vector multiplications!

$$T(\vec{x}) = A_{n \times d} \vec{x}_{d \times 1}$$

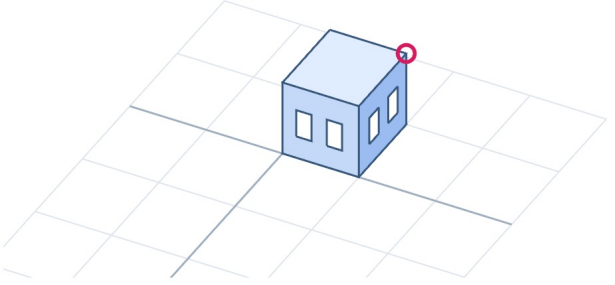
$$T: \underbrace{\mathbb{R}^d}_{\text{domain}} \rightarrow \underbrace{\mathbb{R}^n}_{\text{codomain}}$$

From 3D worlds to 2D screens

View 62% **Building height** 1.50 units

Square grid — Game view | 0 units · flat tile — 2 units

Try game view | Show basis arrows | Reset



$$\begin{pmatrix} 1 \\ 1 \\ 1.50 \end{pmatrix} \xrightarrow{\text{Apply A}} \begin{pmatrix} 64.8 \\ 52.4 \end{pmatrix}$$

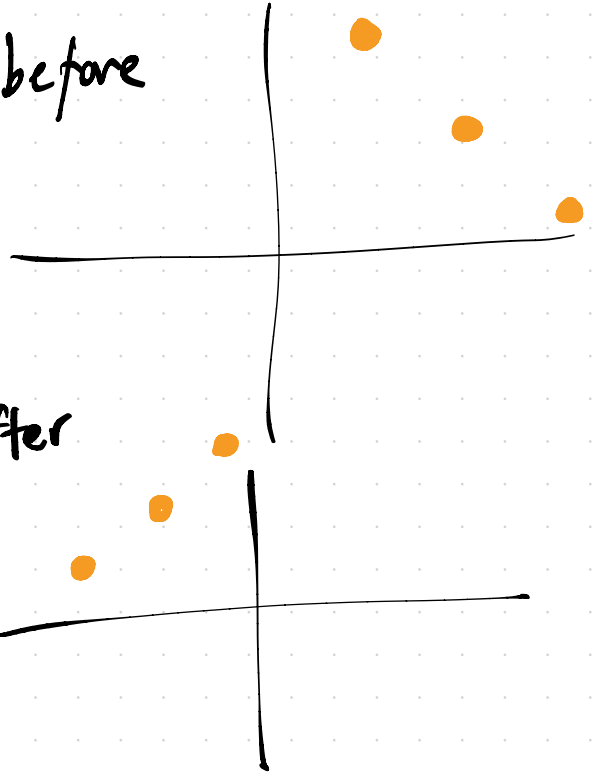
Input in $\mathbb{R}^3$ — Output in $\mathbb{R}^2$

$$A = \begin{bmatrix} 40 & 24.8 & 0 \\ -12.4 & 27.6 & 24.8 \end{bmatrix}$$

Column 1 · x direction | Column 2 · y direction | Column 3 · z direction

The ring marks the roof corner (1, 1, h).

linear transformations preserve straight lines!



Why?

parametric form of line:

position vector

direction vector

$$L: \vec{p} + s\vec{v}, \quad s \in \mathbb{R}$$

what happens if we transform the whole line?

$$T(L): T(\vec{p} + s\vec{v}) = T(\vec{p}) + sT(\vec{v}), \quad s \in \mathbb{R}$$

parametric form of new line

Now: let's focus on linear transformations with a square matrix

e.g. $A_{2\times 2}$

$$T(\vec{x}) = A\vec{x}$$

$$T: \mathbb{R}^2 \to \mathbb{R}^2$$

Mystery

Suppose A 2×2, $T(\vec{x}) = A\vec{x}$

Given $T\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = \begin{bmatrix}15\\4\end{bmatrix} = A\begin{bmatrix}1\\0\end{bmatrix}$

"standard" basis vectors

$T\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}6\\-2\end{bmatrix}, = A\begin{bmatrix}0\\1\end{bmatrix}$

can you find A? $A = \begin{bmatrix}15 & 6\\4 & -2\end{bmatrix}$

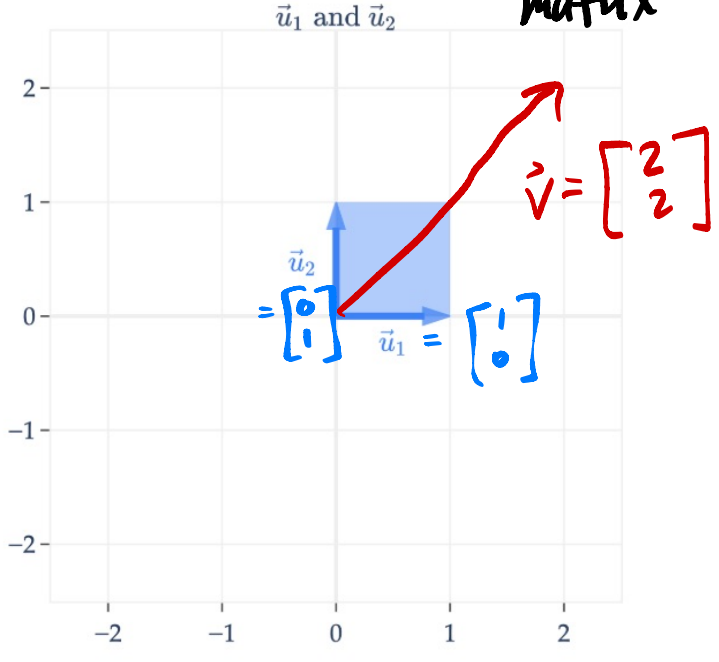
$$A\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 2a \\ -b/3 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 0 \\ 0 & -1/3 \end{bmatrix}$$

diagonal matrix

$$A\vec{v} = \begin{bmatrix} 4 \\ -2/3 \end{bmatrix}$$

A stretches / compresses / reflects each component independently!



$a\vec{u}_1 + b\vec{u}_2 \rightarrow \boxed{T} \rightarrow a\,T(\vec{u}_1) + b\,T(\vec{u}_2)$

$\begin{bmatrix} a \\ b \end{bmatrix}$

with $T(\vec{x}) = \begin{bmatrix} 2 & 0 \\ 0 & -1/3 \end{bmatrix} \vec{x}$,

$$T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = a\,T(\vec{u}_1) + b\,T(\vec{u}_2)$$

$$= a\begin{bmatrix} 2 \\ 0 \end{bmatrix} + b\begin{bmatrix} 0 \\ -1/3 \end{bmatrix}$$

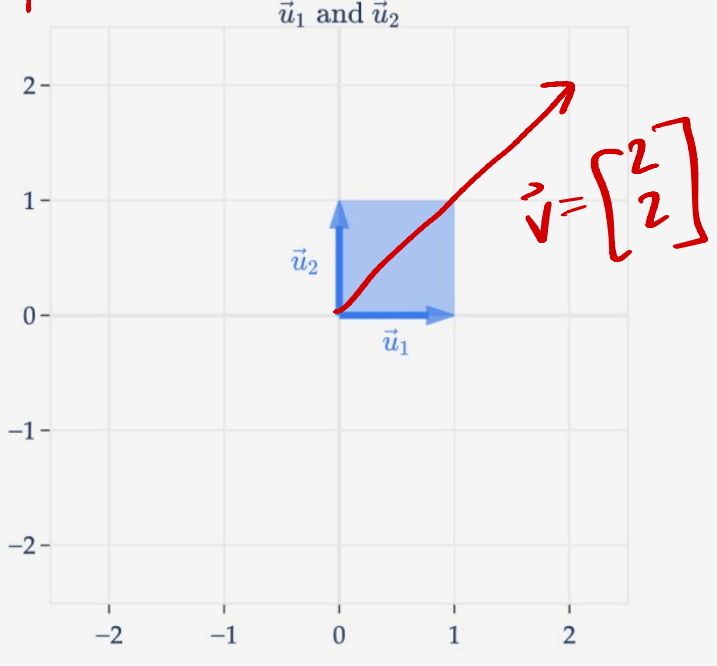
B rotates!

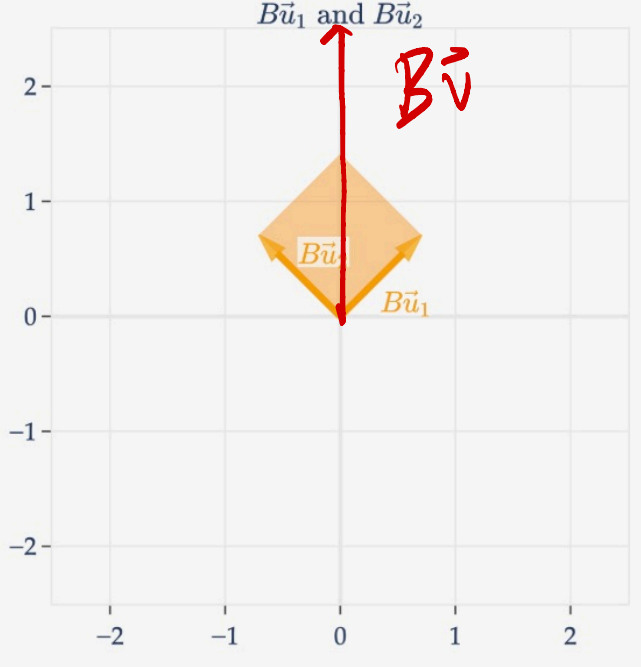
$$\|B\vec{v}\| = \|\vec{v}\|$$

$$B = \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$$

→ orthogonal matrix

$$B^T B = I = BB^T$$





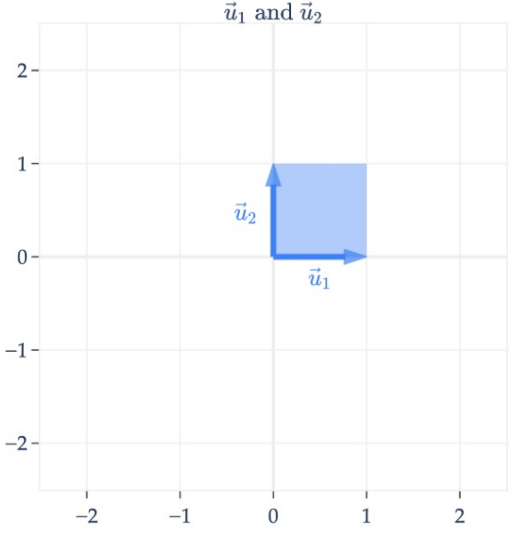
$$C = \begin{bmatrix} \sqrt{2} & -\sqrt{2} \\ -\sqrt{2}/6 & -\sqrt{2}/6 \end{bmatrix}$$

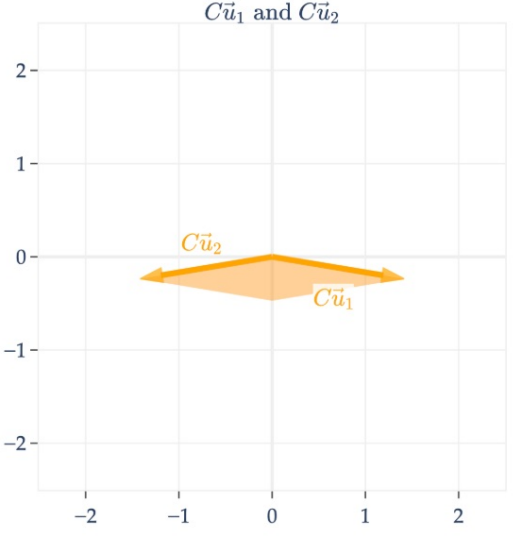
is just

$$C = AB = \underbrace{\begin{bmatrix} 2 & 0 \\ 0 & -1/3 \end{bmatrix}}_{\text{scale}} \underbrace{\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}}_{\text{rotate}} \vec{v}$$

$AB\vec{v}$

rotates, then stretch!

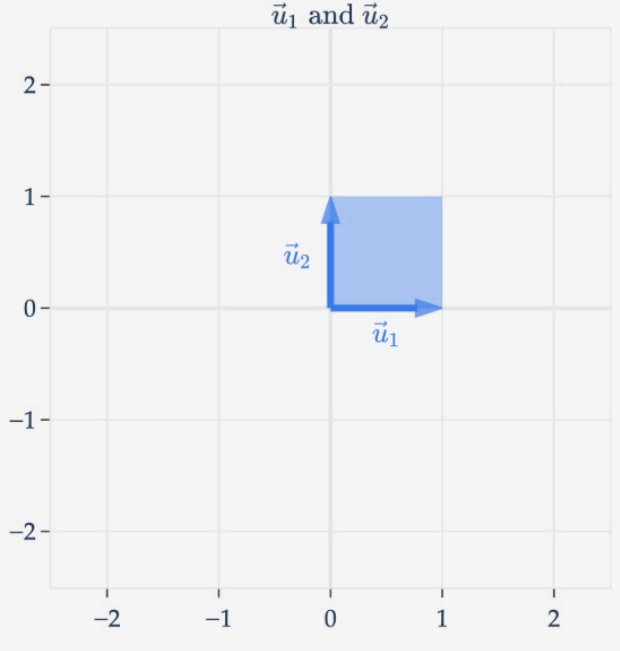


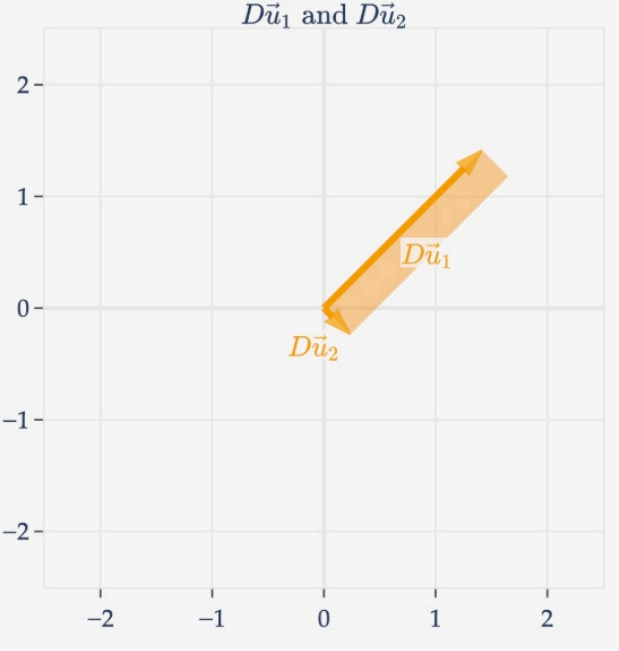


$C = AB$ is different from

$$D = \begin{bmatrix} \sqrt{2} & \sqrt{2}/6 \\ \sqrt{2} & -\sqrt{2}/6 \end{bmatrix} = BA$$

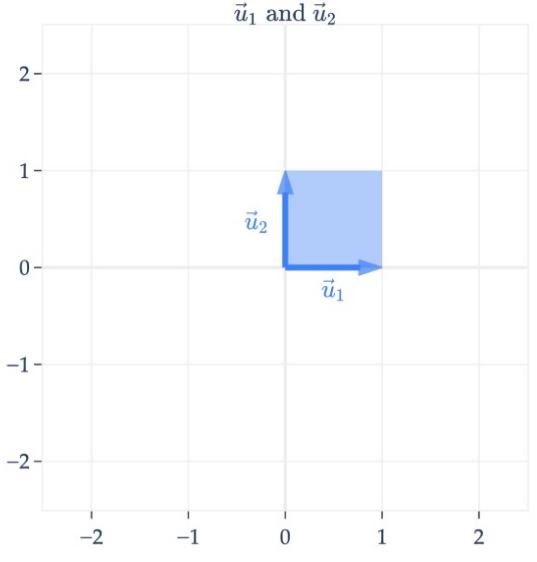
stretch, then rotate!

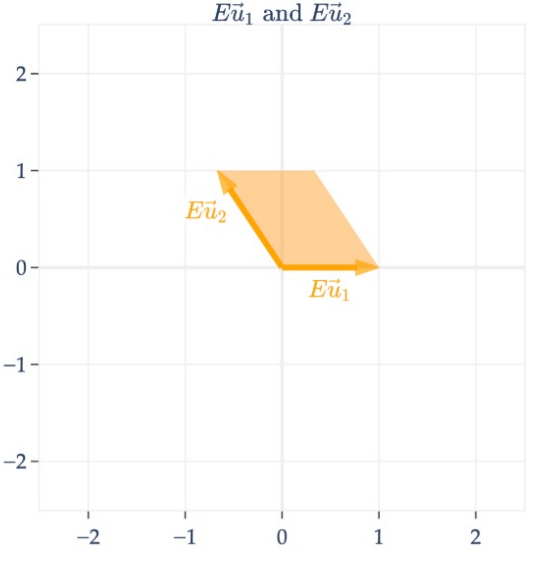




"shear"

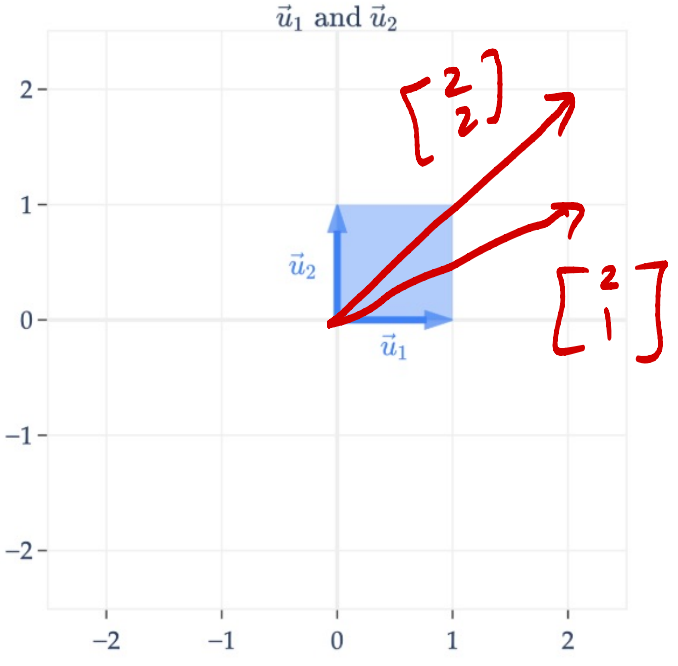
$$E = \begin{bmatrix} 1 & -2/3 \\ 0 & 1 \end{bmatrix}$$

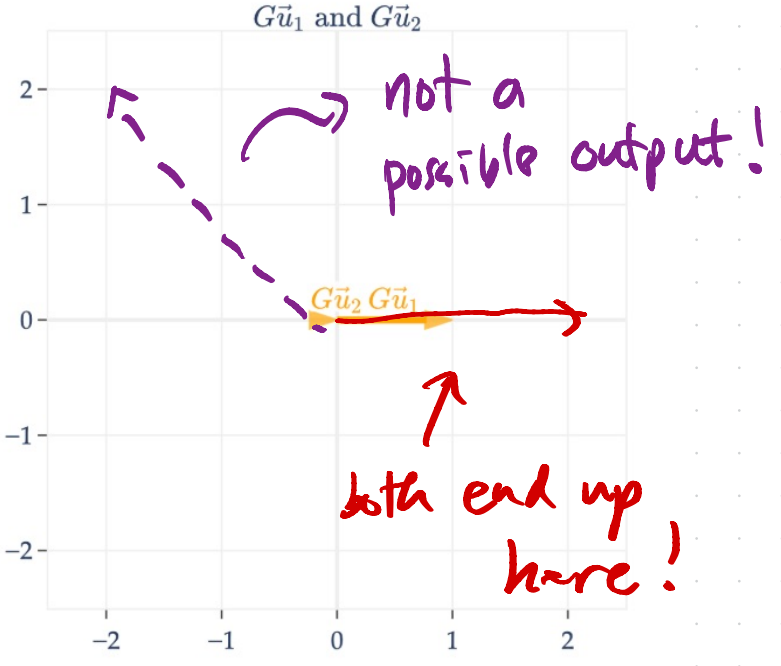




$$G = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}}_{\text{cols not LI}}$$

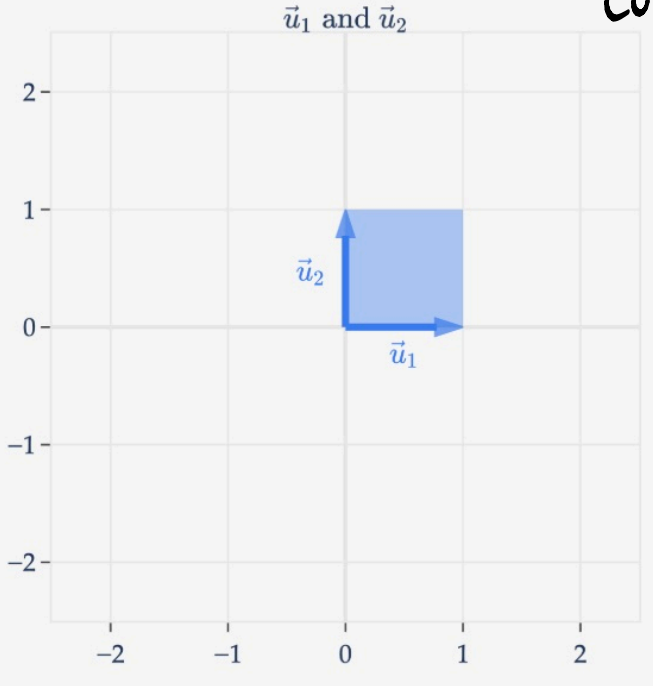
transformation can't be inverted!

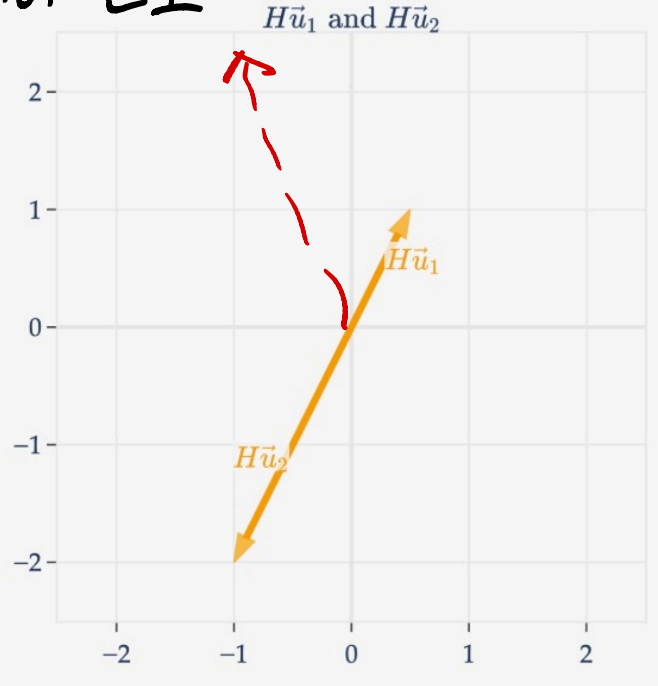




$$H = \begin{bmatrix} 1/2 & -1 \\ 1 & -2 \end{bmatrix}$$

cols not LI





Big idea: A invertible when transformation $T(\vec{x}) = A\vec{x}$ invertible

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Suppose A $n \times n$ matrix. A invertible when:

1. A's cols are linearly independent ($\text{colsp}(A) = \mathbb{R}^n$!)
2. A's rows are linearly independent
3. $\text{rank}(A) = n$
4. $\text{nullsp}(A) = \{\vec{0}\}$
⋮

all equivalent!